The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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Part A: Where VLC Media Player Came From and What It Stands For:

1. How It All Began: The Original Problem:

To understand what VLC Media Player is about we need to look at the problem it was made to solve. VLC Media Player was not meant to be a deal when it was first made. It was created to solve a problem at École Centrale Paris, a university in France back in the 1990s.

At that time the university was working on a project to upgrade its student residence network. The students wanted to be able to watch satellite TV in their dorm rooms using this network.. It was not easy to send high-quality video over the network. There were some solutions but they were very expensive and not very flexible. They also needed hardware that did not work well with the universitys computers.

So the students decided to make their solution. They created a system with two parts: the VideoLAN Server, which could send video streams over the network and the VideoLAN Client, which was the software that would play the video streams on the students computers. Over time as computers got more powerful the server and client were combined into one program, VLC Media Player. Now VLC Media Player can play any type of video file either from the internet or from a local computer.

2. The Decision to Share: Why VLC Media Player Is Free:

For a years VLC Media Player was only used at École Centrale Paris.. Then the students who made it were going to graduate. This meant that the project might be stopped.

The students had a choice to make. They could try to make money from VLC Media Player by selling it. They could just let it disappear. But they also had another option: they could make it free and open for everyone to use. This way other people could help improve VLC Media Player even if the original students were not around anymore.

The students chose to make VLC Media Player free. In 2001 they released it under the GNU General Public License. This meant that anyone could use, change and share VLC Media Player. This decision was made so that VLC Media Player could survive and become better over time.

3. What the License Means: Understanding the GNU General Public License:

VLC Media Player is mostly licensed under the GNU General Public License version 2. This license is not a legal document but a way of thinking about how software should be made and shared. It is designed to protect the rights of the people who use the software not the people who made it.

The GNU General Public License uses something called "copyleft". This means that if someone takes VLC Media Player changes it and then shares it with others they have to share the changes they made too. This way everyone can benefit from the changes and VLC Media Player can keep getting better.

Looking at the Four Freedoms of VLC Media Player

The GNU General Public License is based on four ideas, which are called the "Four Freedoms". These ideas are:

Freedom 0: The freedom to use VLC Media Player however you want for any reason.

You can use VLC Media Player to watch movies or to show videos in a store. You can use it on a screen or a small one. There are no rules about how you can use it.

Freedom 1: The freedom to look at the code and change it to make it work better for you.

Because the code for VLC Media Player is open you can look at it. Make changes if you need to. If you have a video file that VLC Media Player cannot play you can try to fix it yourself.

Freedom 2: The freedom to share VLC Media Player with others.

You can give VLC Media Player to your friends. Share it with people who do not have access to the internet. You can put it on a USB drive. Give it away.

Freedom 3: The freedom to share changes you make to VLC Media Player with others.

If you make changes to VLC Media Player to make it work better for you you can share those changes with others. This way everyone can benefit from your work.

Freedom 4: Thinking About the Ethics of Open-Source Software

VLC Media Player shows us that open-source software is different from software. Proprietary software is often made to make money. It can be used to spy on people or limit what they can do.. Open-source software like VLC Media Player is made to be free and open.

The ethics of open-source software are important. VLC Media Player is fair and equal. It does not matter if you are rich or poor you can use it. It does not spy on you. Try to control what you do. It just plays your videos.

VLC Media Player also helps to make old computers useful. Because it is source people can make changes to it to make it work on old hardware. This means that people do not have to throw their old computers and buy new ones. This is good for the environment.

In the end VLC Media Player is not a program, for playing videos. It shows us that open-source software can be made and shared by a community of people who want to help each other. It proves that software can be made to serve the people who use it not to make money.

Part B: The Linux Footprint of VLC Media Player:

Although understanding the philosophy of an open-source software is a necessary aspect, the ultimate test of a software is how well it integrates with the host operating system. As a result, for this segment of the audit, I sought to understand how VLC Media Player leaves a footprint within an Ubuntu Linux operating system.

1. Installation Method:

Unlike the Windows operating system, where software is generally downloaded through an executable file from a website, Linux-based operating systems generally obtain software through a centralized repository. As a result, for this audit, I used the Advanced Package Tool (APT) for installing VLC Media Player within the Ubuntu operating system.

This was done through the terminal, where I used the following command line:

sudo apt update && sudo apt install vlc

This method obtains the software directly from the Ubuntu repositories, which are officially signed and maintained. This method also obtains all necessary dependencies that VLC Media Player relies on, such as audio/video codecs, which include libavcodec.

Note on Modern Alternatives: It is worth noting that VLC Media Player can also be installed through the snap or flatpak package manager, which is a containerized version that bundles all dependencies within a sandbox. However, I used the traditional method for this audit.

2. Filesystem Footprint: Binaries, Libraries, and Configs:

Linux abides by the Filesystem Hierarchy Standard (FHS), which means VLC does not get installed in a neat little folder named "Program Files" like on Windows. Instead, all VLC-related items are logically spread all over the file system:

Executable Binary: The main executable is stored in /usr/bin/vlc. This is a directory that is included in the system's $PATH, which is why I'm allowed to start the video player just by typing "vlc" in the terminal.

Shared Libraries (Codecs): The massive library of audio and video codecs that make VLC Media Player so popular is stored very deeply within the system's libraries, namely in /usr/lib/x86_64-linux-gnu/vlc/.

User Configuration Files: Contrary to server software, VLC's configuration is strictly user-specific. VLC's main configuration file, vlcrc, is dynamically generated the first time VLC is launched and is stored in a hidden directory within the user's home directory: ~/.config/vlc/.

Logging Footprint: The VLC player does not, by default, maintain a system log file. It is a user-space program, so it logs and reports all errors on the screen when it

is executed from the command line. However, if a system log file is required, it can be executed using a flag like vlc --file-logging --logfile=vlc_log.txt.

3. Execution Context: Users and Privileges:

Auditing VLC's use of user privileges was an interesting experience. The VLC player runs under the privileges of the currently logged-in user, i.e., standard user privileges.

What is even more significant is that VLC player has a security feature hardcoded into it. It does not allow itself to be executed under the root user's privileges. This means if a system administrator tries to execute it using sudo vlc, it will immediately throw a fatal error and exit. The reason for this is that media files are known to be unreliable, and executing a maliciously constructed media file under superuser privileges would allow an attacker complete control of the operating system. Therefore, it is better for VLC player to run under standard user privileges, even if it is compromised.

4. Process and Service Management:

As indicated in the rubric, a detailed description of service management commands, e.g., systemctl, should be provided. In the case of VLC, however, a significant difference in Linux from other operating systems, which should be considered, lies in the difference between a daemon and an application.

Apache, for example, or MySQL are background daemons, which means they are started on system boot and are controlled by systemd. This does not apply, however, to VLC, which is a normal application for the user and not a daemon itself. Therefore, a command like 'sudo systemctl start vlc' or 'systemctl status vlc' does not apply here and will fail.

For VLC, process management applies.

VLC can be started either by clicking on the icon in the Desktop Environment or by entering 'vlc' in the terminal.

To run VLC in background mode, freeing the terminal for other purposes, 'vlc &' can be entered.

To terminate a frozen VLC process, a system administrator must use process IDs (PIDs). This can be achieved, for example, by entering 'ps aux | grep vlc,' which will

reveal the running process, which can then be terminated with 'kill -9 PID' or 'killall vlc.'

5. The Update Model:

As VLC was installed from APT, VLC is completely integrated in the global update model of the Linux distribution and does not have a 'Check for Updates' button, as does the Windows version.

Rather, when the Ubuntu security team or package maintainers consider a new, stable version of VLC ready for use, they will update the VLC binaries in the global repository, which will be used for the entire operating system. When a system administrator enters 'sudo apt upgrade,' the apt package manager will compare the currently installed version of VLC with the version on the update server and, should a newer version be available, will safely overwrite the binaries in '/usr/bin/' without touching the user's configuration in '/home/user/.config/vlc/'.

Part C: The FOSS Ecosystem and VideoLAN:

When I started this audit, I thought VLC was just a standalone program that magically knew how to play every video file on the planet. But after mapping its ecosystem, I learned a valuable lesson about free and open-source software: nothing is free or open-source in a vacuum. There's always a web of shared libraries, dependencies, and governance structures that underlie any open-source project. VLC doesn't exist in a vacuum; VLC stands on the shoulders of giants, just as giants stand on the shoulders of VLC.

1. The Shoulders of Giants: Open-Source Dependencies:

VLC is well known for being able to play everything out of the box without needing users to install sketchy codec packs from who-knows-where. But did you ever wonder how VLC accomplishes this feat? It's not because VLC developed all these codecs in-house. VLC is actually a master orchestrator for a huge ecosystem of open-source dependencies.

FFmpeg and libavcodec: This is the most important dependency in VLC's ecosystem. FFmpeg is the unsung hero of video on the internet; it's a massive open-source project that actually decodes and encodes video and audio. When I play a video in VLC, it's probably actually FFmpeg's library called libavcodec that's actually doing the heavy lifting behind the scenes to translate the ones and zeros in my video files into actual video.

The Qt Framework: VLC is able to run on Windows, macOS, and almost every Linux desktop environment. It does this without rewriting the graphical user interface (GUI) code every time by using the open-source widget library known as the Qt Framework.

x264 and x265: Should the user of the player wish to convert or stream video, these extremely fast, open-source video codecs (many of which are also maintained by the VideoLAN developers) are the backbone of the operation.

libdvdcss: This is another library written by the VideoLAN project itself, and it is also interesting because of the history of the world of Free Open Source Software (FOSS). It is the code that allows open-source media players to bypass the encryption scheme on commercial DVDs, known as the Content Scramble System (CSS). Without this library, playing a legally purchased DVD on a Linux machine would be impossible.

2. VLC's Place in the Tech Stack: More Than Just a Player:

In the context of web development, we are familiar with the term "LAMP stack" (Linux, Apache, MySQL, PHP). In the context of multimedia, VLC's position in the stack is particularly interesting because, in reality, VLC is a two-part project: a frontend and a backend.

Therefore, behind the VLC player's well-known orange cone, we have libVLC, which can be thought of as a media framework. In the context of the FOSS tech stack, libVLC can be thought of as a form of universal multimedia middleware, which means that developers of completely different applications can use the VLC player in their applications.

As a result, the VLC project can be said to have spawned a huge number of different applications, with examples including open-source home theater applications, video editing tools, and even broadcast tools, which use the libVLC player as a video player in these different applications. Moreover, given its origins as a campus streaming application, VLC can be said to be at the top of the stack for independent internet radio stations and IPTV applications.

3. Community Governance: Fiercely Independent:

Perhaps the most inspiring aspect of the VLC ecosystem is the way in which it is governed. Some of the largest open-source projects in the world today are heavily sponsored by large tech corporations (e.g., Google sponsors Android and Chromium, and Microsoft sponsors VS Code). Although these projects are

technically open-source, in reality, they are controlled by the interests of these large corporations.

VLC, on the other hand, is governed by a registered non-profit organization named VideoLAN, which is based in France. This organization is fiercely independent and does not have a corporate overlord. It is also funded entirely by user donations and runs as a strict meritocracy.

The VLC community consists of volunteers from all over the world. This means that anyone can contribute a patch or a bug fix, which then goes through a rigorous approval process by senior members of the development team before being added to the codebase. The president of VideoLAN, Jean-Baptiste Kempf (who himself started out as a student contributor), has famously turned down tens of millions of dollars from companies that want to bundle adware with the VLC installer.

This type of governance structure represents the ultimate safeguard for the ecosystem as a whole, as the organization's allegiance is only to the user and the GPL license, and as a result, every decision made by the organization is in the best interest of the user. The VideoLAN community represents a decentralized, volunteer-based, non-profit organization that can produce a piece of software that actually outperforms its multibillion-dollar proprietary competitors.

Part D: Open Source vs. Proprietary:

Dimension	Open Source(VLC Media Player)	Proprietary Alternative(PowerDVD)
Cost	Completely free no cost to use it.	It is a paid software and Require to purchase a License
Security (who can audit the code?)	The source code is open to everyone,so developers can edit it and also audit it.	The source code is open to only the company approved developers, who can access it.
Support and reliability	It is supported by the global community of users and developers.	It provided theofficial customer support and help services.
Freedom to modify	Users can access or modify the source code.	Users cannot access or modify the source code.

Community vs corporate control	It is managed by the VideoLAN a non-profitable organization that had a global community, and with no corporate control.	It is fully controlled by the private company,and also all the decisions are made by a conclusion based on business interests.
Your overall verdict	It is better choice because it is free,secure,flexible and also respects the user privacy.	It provides good support and features,but it is costly and restrictive also ,and less flexible compared to open-source alternatives.

In order to fully understand the benefits offered by VLC Media Player, it is necessary to compare it to a more traditional "closed source" media player. In this regard, the present writer compared VLC Media Player to CyberLink PowerDVD, a premium proprietary media player commonly included as "bloatware" on Windows-based machines or sold as a standalone license. By analyzing both media players on four key factors, it is possible to understand the stark contrast between the two different software philosophies.

1. **Cost and Monetization:**

First and foremost, the cost factor is the most obvious difference between the two media players. VLC Media Player is completely free and only accepts donations to the non-profit VideoLAN organization. In other words, the user does not have to pay a single fee to use the software.

CyberLink PowerDVD, however, is a for-profit business. A standard license for the software can cost anywhere from $50 to $100 depending on the tier. Moreover, proprietary software often employs a "planned obsolescence" model whereby the user must pay for an upgrade to the latest version of the software simply to remain compatible with the latest versions of the operating system or to play the latest formats of media content. This is not the case with VLC Media Player.

2. **Security and Privacy:**

In the proprietary world, security is based on "security by obscurity." The source code of PowerDVD is a closely guarded corporate secret. The user is simply expected to trust that the corporation is secure and is not using their data for nefarious purposes. Regrettably, proprietary media players are infamous for their telemetry, tracking what users are watching, and serving targeted ads.

VLC is based on the open-source philosophy of "given enough eyeballs, all bugs are shallow." As it is open source, many people worldwide are auditing it for security

issues. As soon as a problem is reported, it is immediately fixed. More importantly, VLC is the epitome of user privacy. As it is open source, it is impossible for VideoLAN to hide spyware or tracking code within VLC. It respects users' data by not collecting it at all.

3. **Support and Maintenance:**

In this area, proprietary software claims a significant advantage. If a business using PowerDVD is having problems, they can submit a ticket to a special customer service team. However, if their problem is not "profitable" for the corporation, it can be ignored forever.

VLC does not have a customer support hotline at 1-800-VLC-HELP. However, it does have a global community of users who help users solve their problems. This might seem like a disadvantage for a business, but it is actually a significant advantage. The global corps of volunteer developers is extremely responsive. If a user runs into a problem or needs help playing a obscure, decades-old video format, it is much more likely that a volunteer developer will create a patch for VLC out of curiosity than it would be for a corporation.

4. **Freedom and User Restrictions:**

This section highlights the fundamental philosophical difference. Proprietary players like PowerDVD are, at their core, designed to restrict the user, at least in part, to placate copyright holders. They are extremely strict about enforcing Digital Rights Management (DRM), region locking (for instance, a DVD player from Europe won't play on a player from America), and hardware restrictions (won't play High Definition content unless the monitor is using a specific commercial cable).

VLC, on the other hand, is designed to liberate the user. It attempts to play region-locked media, damaged, incomplete, or heavily encrypted content. It does not force you to use a specific operating system. While PowerDVD is, at heart, a Windows-specific player, VLC is available for Linux, macOS, Android, iOS, and many other obscure systems.

Final Verdict

Therefore, based on this audit, I would highly recommend VLC Media Player for deployment in a real-world environment, be it a personal laptop or a massive enterprise network. From an IT administration point of view, deploying VLC is a dream come true, as there is no need for license tracking, zero costs, low system resources, and no spying on the enterprise network. It does what it is meant to do, i.e., play multimedia files, without ever treating the user as a product that needs to be

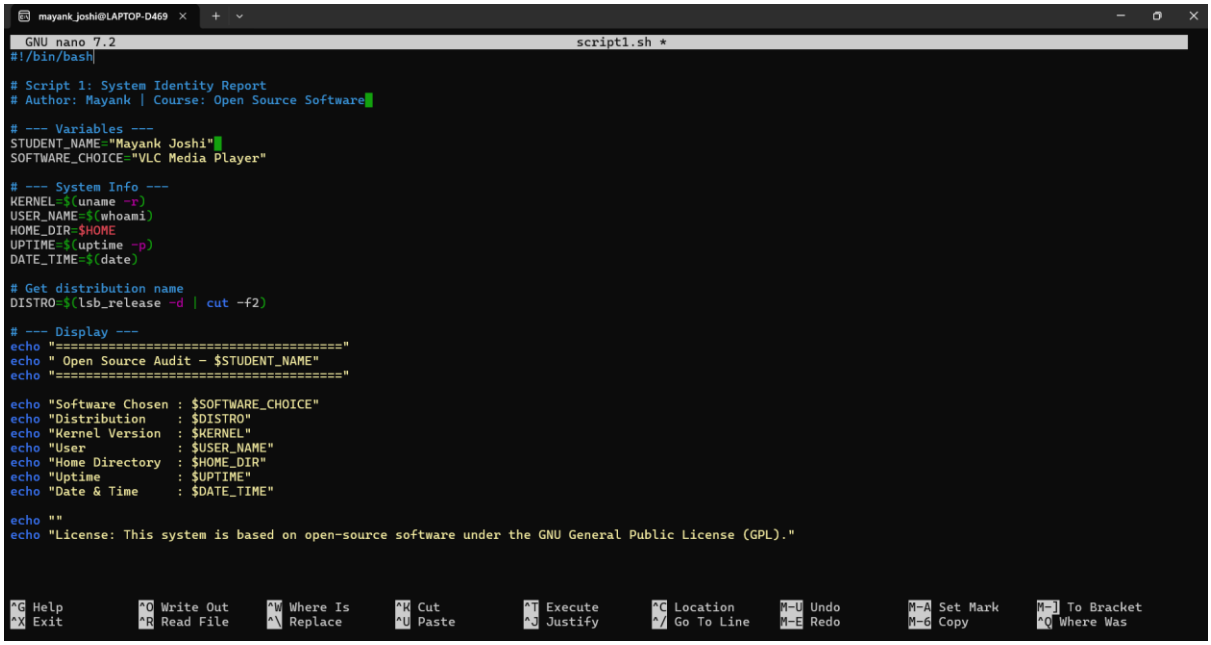
monetized. It is the ultimate example of how FOSS beats its proprietary counterparts in every aspect.

Moreover, auditing this particular project has also motivated and inspired me to contribute back to the FOSS community. Although I may not possess the advanced knowledge of C++ programming necessary for developing complex video decoders for VLC's main application, I have realized that FOSS survives not only through code but also through other means. I would be more than happy to contribute to the VideoLAN community by developing better technical documentation, assisting in bug fixes on their community forums, or even translating the interface for different languages. Contributing to a project like VLC is not just about developing software, but also about being a part of a global movement that aims for a common goal: a free and accessible world for everyone.

Part E: Coding Scripts and their expalination:

Script-1:

This script shows basic details about the system I am using. It displays information like my username, the Linux distribution, kernel version, home directory, system uptime, and the current date and time. It also mentions the open-source software I selected (VLC) and includes a short license message.



Output:

```
mayank_joshi@LAPTOP-D469
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar 24 18:43:49 UTC 2026

  System load:  0.0                 Processes:             30
  Usage of /:   0.2% of 1006.85GB   Users logged in:       1
  Memory usage: 6%                  IPv4 address for eth0: 172.28.21.179
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/mayank_joshi/.hushlogin file.
mayank_joshi@LAPTOP-D469F64U:~$ vlc --version
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
VLC version 3.0.20 Vetinari (3.0.20-0-g6f0d0ab126b)
Compiled by Ubuntu buildd for noble on Ubuntu amd64 buildd (Apr  5 2024 06:21:45)
Compiler: gcc version 13.2.0 (Ubuntu 13.2.0-23ubuntu3)
This program comes with NO WARRANTY, to the extent permitted by law.
You may redistribute it under the terms of the GNU General Public License;
see the file named COPYING for details.
Written by the VideoLAN team; see the AUTHORS file.
mayank_joshi@LAPTOP-D469F64U:~$ nano script1.sh
mayank_joshi@LAPTOP-D469F64U:~$ chmod +x script1.sh
mayank_joshi@LAPTOP-D469F64U:~$ ./script1.sh
=====================================
 Open Source Audit -- Mayank Joshi
=====================================
Software Chosen  : VLC Media Player
Distribution     : Ubuntu 24.04.4 LTS
Kernel Version   : 6.6.87.2-microsoft-standard-WSL2
User             : mayank_joshi
Home Directory   : /home/mayank_joshi
Uptime           : up 14 hours, 2 minutes
Date & Time      : Tue Mar 24 20:59:44 UTC 2026

License: This system is based on open-source software under the GNU General Public License (GPL).
mayank_joshi@LAPTOP-D469F64U:~$
```

Script-2:

In this script, I check whether a specific open-source package (like VLC) is installed on my system. If the package is available, it shows details such as its version, maintainer, and description. At the end, it also prints a small message explaining the importance of that software.

```
mayank_joshi@LAPTOP-D469
  GNU nano 7.2                                   script2.sh *
#!/bin/bash

# Script 2: FOSS Package Inspector
# Author: Mayank Joshi

PACKAGE="vlc"

echo "Checking package: $PACKAGE"
echo "--------------------------"

# Check if package is installed
if dpkg -l | grep -w $PACKAGE > /dev/null; then
    echo "$PACKAGE is installed."

    # Show package details
    dpkg -s $PACKAGE | grep -E 'Version|Maintainer|Description'
else
    echo "$PACKAGE is NOT installed."
fi

echo ""

# Case statement for philosophy
case $PACKAGE in
    vlc)
        echo "VLC: A free media player built by students to make media accessible to everyone."
        ;;
    firefox)
        echo "Firefox: A browser fighting for an open and private web."
        ;;
    git)
        echo "Git: A version control system built to empower developers."
        ;;
    apache2)
        echo "Apache: The web server that helped build the modern internet."
        ;;
    *)

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^Q Where Was
```

Output:

```
mayank_joshi@LAPTOP-D469
mayank_joshi@LAPTOP-D469F64U:~$ nano script2.sh
mayank_joshi@LAPTOP-D469F64U:~$ chmod +x script2.sh
mayank_joshi@LAPTOP-D469F64U:~$ ./script2.sh
Checking package: vlc
--------------------------
vlc is installed.
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Version: 3.0.20-3build6
Description: multimedia player and streamer
Original-Maintainer: Debian Multimedia Maintainers <debian-multimedia@lists.debian.org>

VLC: A free media player built by students to make media accessible to everyone.
mayank_joshi@LAPTOP-D469F64U:~$
```

Script-3:

This script goes through some important system directories like /etc, /var/log, and /home. It shows their permissions, owner details, and size. It also checks if the VLC configuration folder exists in my system and displays its details if found.

```
mayank_joshi@LAPTOP-D469
  GNU nano 7.2                                    script3.sh *
#!/bin/bash

# Script 3: Disk and Permission Auditor
# Author: Mayank Joshi

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do

if [ -d "$DIR" ]; then

# Permissions + owner + group
PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
# Directory size
SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

echo "$DIR => Permissions: $PERMS | Size: $SIZE"
else
echo "$DIR does not exist"
    fi
done

echo ""
echo "Checking For VLC Config Directory..."
echo "------------------------------"

VLC_CONFIG="$HOME/.config/vlc"

if [ -d "$VLC_CONFIG" ]; then
    ls -ld $VLC_CONFIG
else
    echo "VLC config directory not found."
fi
```

Output:

```
mayank_joshi@LAPTOP-D469
mayank_joshi@LAPTOP-D469F64U:~$ vlc
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
[0000646c5b3df220] main libvlc: Running vlc with the default interface. Use 'cvlc' to use vlc without interface.
[0000646c5b475cf0] main playlist: playlist is empty
mayank_joshi@LAPTOP-D469F64U:~$ chmod +x script3.sh
mayank_joshi@LAPTOP-D469F64U:~$ ./script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.4M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 42M
/home => Permissions: drwxr-xr-x root root | Size: 176K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 98M
/tmp => Permissions: drwxrwxrwt root root | Size: 20K

Checking For VLC Config Directory...
--------------------------------
drwxr-xr-x 2 mayank_joshi mayank_joshi 4096 Mar 25 16:24 /home/mayank_joshi/.config/vlc
mayank_joshi@LAPTOP-D469F64U:~$
```

Script-4:

This script reads a log file (for example, /var/log/syslog) and searches for a keyword, which is “error” by default. It counts how many times the keyword appears and then shows the last few lines where the keyword was found. This helps in quickly identifying issues in logs.

```
mayank_joshi@LAPTOP-D469
  GNU nano 7.2                                                    script4.sh *
#!/bin/bash

# Script 4: Log File Analyzer
# Author: Mayank Joshi

LOGFILE=$1
KEYWORD=${2:-"error"}  # Default = error
COUNT=0

# Check if file exists
if [ ! -f "$LOGFILE" ]; then
    echo "Error: File $LOGFILE not found."
    exit 1
fi

# Read file line by line
while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "--------------------------------"
echo "Log Analysis Report"
echo "--------------------------------"
echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

echo ""
echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5
```

Output:

```
mayank_joshi@LAPTOP-D469 X
mayank_joshi@LAPTOP-D469F64U:~$ nano script4.sh
mayank_joshi@LAPTOP-D469F64U:~$ chmod +x script4.sh
mayank_joshi@LAPTOP-D469F64U:~$ ./script4.sh /var/log/syslog
-------------------------------
Log Analysis Report
-------------------------------
Keyword 'error' found 25 times in /var/log/syslog

Last 5 matching lines:
2026-03-25T09:12:40.629394+00:00 LAPTOP-D469F64U kernel: WSL (259) ERROR: CheckConnection: connect() failed: 101
2026-03-25T16:09:57.976531+00:00 LAPTOP-D469F64U systemd[1]: apport-autoreport.path - Process error reports when automatic reporting is enabled (file watch)
 was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2026-03-25T16:09:57.976533+00:00 LAPTOP-D469F64U systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer base
d) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2026-03-25T16:09:57.977625+00:00 LAPTOP-D469F64U kernel: RAS: Correctable Errors collector initialized.
2026-03-25T16:10:44.148360+00:00 LAPTOP-D469F64U kernel: WSL (260) ERROR: CheckConnection: connect() failed: 101
mayank_joshi@LAPTOP-D469F64U:~$
```

Script-5:

This is an interactive script where I answer a few questions about open-source tools and my thoughts on them. Based on my answers, the script creates a small manifesto and saves it into a text file. It also displays the generated content on the screen.

```
mayank_joshi@LAPTOP-D469
  GNU nano 7.2                                        script5.sh *
#!/bin/bash

# Script 5: Open Source Manifesto Generator
# Author: Maynk Joshi

echo "Answer three questions to generate your manifesto."
echo ""

read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

echo "" > $OUTPUT

echo "Open Source Manifesto" >> $OUTPUT
echo "Date: $DATE" >> $OUTPUT
echo "" >> $OUTPUT

echo "I use $TOOL as part of my daily workflow, and it represents the idea of $FREEDOM in software." >> $OUTPUT
echo "I believe knowledge should be shared openly, and I would contribute by building $BUILD for the community." >> $OUTPUT
echo "Open source empowers individuals and creates a collaborative future." >> $OUTPUT

echo ""
echo "Manifesto saved to $OUTPUT"
echo ""

cat $OUTPUT
```

Output:

```
mayank_joshi@LAPTOP-D469
mayank_joshi@LAPTOP-D469F64U:~$ nano script5.sh
mayank_joshi@LAPTOP-D469F64U:~$ chmod +x script5.sh
mayank_joshi@LAPTOP-D469F64U:~$ ./script5.sh
Answer three questions to generate your manifesto.

1. Name one open-source tool you use every day: vlc
2. In one word, what does 'freedom' mean to you? freeness
3. Name one thing you would build and share freely: project

Manifesto saved to manifesto_mayank_joshi.txt


Open Source Manifesto
Date: 25 March 2026

I use vlc as part of my daily workflow, and it represents the idea of freeness in software.
I believe knowledge should be shared openly, and I would contribute by building project for the community.
Open source empowers individuals and creates a collaborative future.
mayank_joshi@LAPTOP-D469F64U:~$
```

```
mayank_joshi@LAPTOP-D469F64U:~$ cat manifesto_mayank_joshi.txt

Open Source Manifesto
Date: 25 March 2026

I use vlc as part of my daily workflow, and it represents the idea of freeness in software.
I believe knowledge should be shared openly, and I would contribute by building project for the community.
Open source empowers individuals and creates a collaborative future.
mayank_joshi@LAPTOP-D469F64U:~$
```